

ClassRoomSTORM Theory Notes

Diffraction, Blinking, and the Physics of Localization Microscopy

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Companion to ClassRoomSTORM Studio, software version 1.2

Abstract

These notes explain the physics behind ClassRoomSTORM Studio V1.2. They proceed from the diffraction limit of an ordinary microscope to the idea of localization microscopy: why a blurred spot can still be *located* far more precisely than it can be *resolved*, and how many such locations accumulate into a sharp image. The central distinction is between two quantities: *resolution* (how finely an instrument separates two points in a single snapshot) is set by optical blur, whereas *localization precision* (how well the centre of one isolated spot is known) is a statistical quantity set by photon counts, background, and noise. The final sections describe how ClassRoomSTORM V1.2 reconstructs an image, and explain the capabilities and limitations of the centroid method. The V1.2 release adds calibrated coordinate export, crop-aware truth validation, fractional-coordinate rendering, and optional frame-median subtraction while retaining the straightforward centroid reconstruction method.

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1 Introduction: The Resolution Problem

A microscope cannot show arbitrarily fine detail. Two points of light that are close enough together blur into a single spot, even with an ideal lens. This limit, the *diffraction limit*, has been understood since the nineteenth century [1] and is roughly half the wavelength of light, about 200-250 nm for visible light [21, 13].

For more than a century this was considered an unbreakable barrier. Modern *super-resolution* microscopy sidesteps it. One family of methods, *single-molecule localization microscopy* (SMLM), which includes STORM and PALM, uses a simple statistical principle: never let two nearby emitters be visible at the same time. If only a few, well-separated emitters glow in any one frame, each appears as an isolated blurred spot, and the *centre* of an isolated spot can be estimated very precisely. Repeating this over thousands of frames and accumulating the centres yields a far sharper picture than any single frame [16, 15].

ClassRoomSTORM is a teaching application that reproduces the *statistical and computational logic* of SMLM. Its *Virtual Experiment* generates blinking videos with controllable blur, brightness, and noise; its *SR Reconstruction* workflow locates the spots and accumulates them. These notes explain the physics underlying the software. They are meant to be read alongside the *ClassRoomSTORM Help* document, which explains the buttons and screens.

A note on scope

ClassRoomSTORM is a learning tool. It is not a fluorescence microscope, and its reconstruction uses a deliberately simple estimator rather than a research-grade one. Throughout these notes, material describing the *ideal* statistical method is clearly marked in violet boxes; the running text and section 10 describe what ClassRoomSTORM V1.2 actually does.

2 The Diffraction Limit

When light from a single point passes through a lens of finite size, diffraction spreads it into a small pattern of rings called the *Airy pattern*:

$$I(r) = I_0 \left[\frac{2J_1(\alpha r)}{\alpha r} \right]^2, \quad \alpha = \frac{2\pi \text{NA}}{\lambda}, \quad (1)$$

where r is the radial distance from the centre, I_0 is the peak intensity, J_1 is the first-order Bessel function, λ is the wavelength of light, and NA is the *numerical aperture* of the lens (a measure of how wide a cone of light it collects). A perfect point of light is therefore never imaged as a point; it appears as a small bright disc.

Two points can be distinguished only if their discs do not overlap too much. Abbe's criterion gives the smallest resolvable separation [1, 21]:

$$d_{\text{Abbe}} \simeq \frac{\lambda}{2 \text{NA}}. \quad (2)$$

A closely related statement, the Rayleigh criterion, places the threshold where the peak of one Airy disc falls on the first dark ring of the other [21, 18]:

$$d_{\text{Rayleigh}} \simeq \frac{0.61 \lambda}{\text{NA}}. \quad (3)$$

Worked example

Consider green light with $\lambda \approx 550$ nm, and an objective with $\text{NA} \approx 1.4$. Then $d_{\text{Abbe}} \approx 550/(2 \times 1.4) \approx 196$ nm and $d_{\text{Rayleigh}} \approx 0.61 \times 550/1.4 \approx 240$ nm. This is the few-hundred-nanometre barrier that super-resolution methods must overcome.

Figure 1 shows how two emitters merge as their separation shrinks toward this limit. The key point is that eqs. (2) and (3) describe *resolution*, the ability to separate two points that are visible *at the same time*. They say nothing about how well the position of a *single, isolated* point can be measured. A single point can be located much more precisely than two points can be resolved.

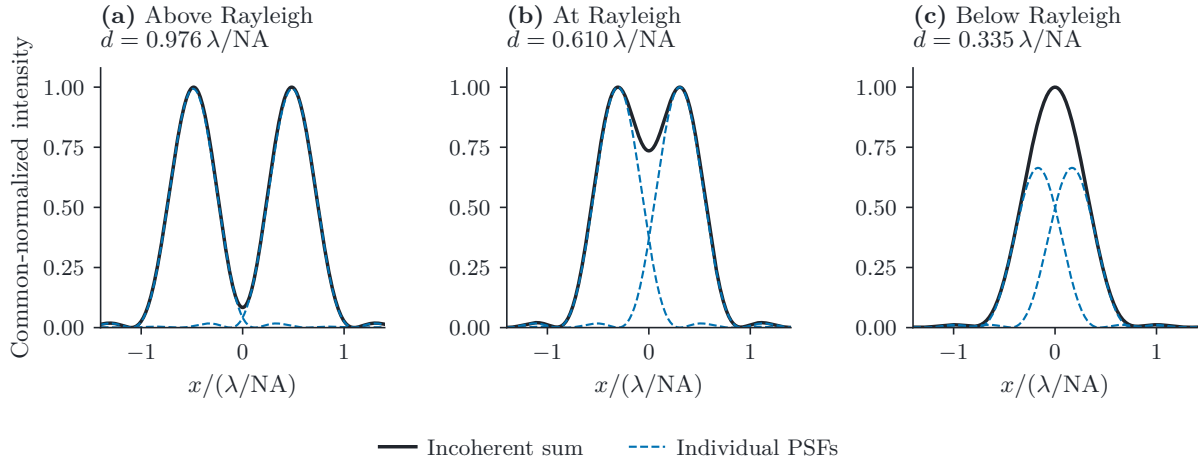


Figure 1: Two incoherent point sources of equal brightness imaged through an ideal circular pupil. Each panel uses the same normalization for the sum and both component PSFs, so the components add to the plotted sum. Rayleigh separation is $0.610 \lambda/\text{NA}$; it is a conventional two-point criterion, not a universal localization limit.

3 The Point Spread Function

The image of a single point source is called the *point spread function* (PSF). The Airy pattern of eq. (1) is the exact PSF of an ideal lens, but for calculations a simpler model is almost always used: a two-dimensional Gaussian [6, 10]. An isolated emitter is modelled as

$$I(x, y) = b + \frac{N}{2\pi\sigma^2} \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2} \right], \quad (4)$$

where (x_0, y_0) is the true emitter position, N is the total number of signal photons, b is a background level, and σ is the *width* of the PSF. The width is set by the optics and the pixel size a of the camera; a useful approximate relation is [6]

$$\sigma \approx 0.21 \frac{\lambda}{\text{NA} \cdot a}. \quad (5)$$

The Gaussian is not exact because it ignores the Airy rings, but it captures the central peak well (fig. 2) and makes the algebra of localization tractable.

In ClassroomSTORM, the *Virtual Experiment* renders each blinker as a Gaussian spot, and you set its width directly with the `sigma_px` control. Choosing a large σ is the same as building a more strongly diffraction-limited “microscope.”

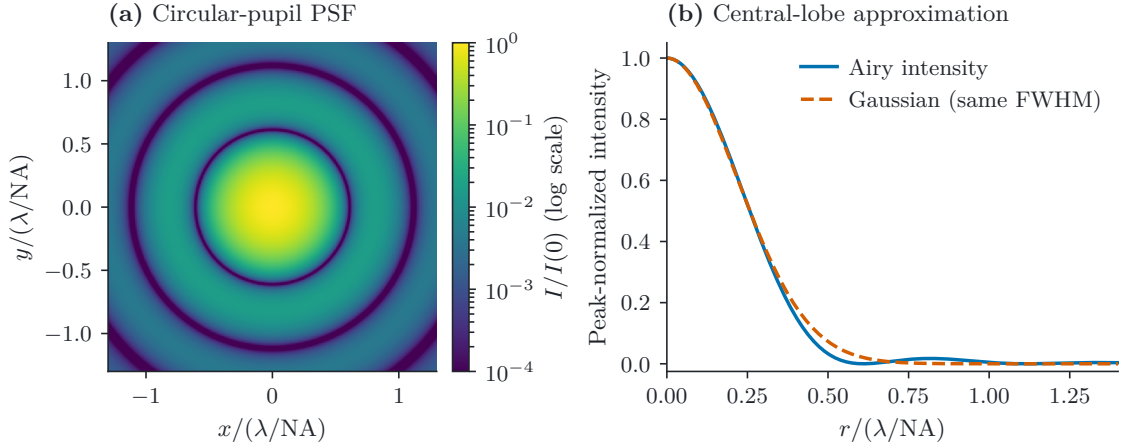


Figure 2: Ideal scalar, incoherent circular-pupil PSF. The image uses a labelled logarithmic colour scale to reveal rings; the profile is linear. The Gaussian matches the Airy FWHM only, not the side lobes or total energy.

4 Image Formation and Sampling

A real image is the true scene blurred by the PSF and then corrupted by noise. Writing the true distribution of light as $f(x, y)$, the PSF as $h(x, y)$, and the noise as $\eta(x, y)$, the recorded image is

$$g(x, y) = (h * f)(x, y) + \eta(x, y), \quad (6)$$

where $*$ denotes convolution. Convolution by h describes the blurring discussed in section 2: every point in f is replaced by a copy of the PSF, as fig. 3 illustrates.

A camera does not sample g at infinitely fine points. Each pixel *collects all the light that lands on its small square*. The expected number of counts in the pixel at column j , row i , for a single Gaussian emitter, is the Gaussian *integrated* over that pixel [7]:

$$M_{ij} = b + N G_x(j; x_0, \sigma) G_y(i; y_0, \sigma), \quad (7)$$

$$G_x(j; x_0, \sigma) = \frac{1}{2} \left[\operatorname{erf} \frac{j + \frac{1}{2} - x_0}{\sqrt{2} \sigma} - \operatorname{erf} \frac{j - \frac{1}{2} - x_0}{\sqrt{2} \sigma} \right], \quad (8)$$

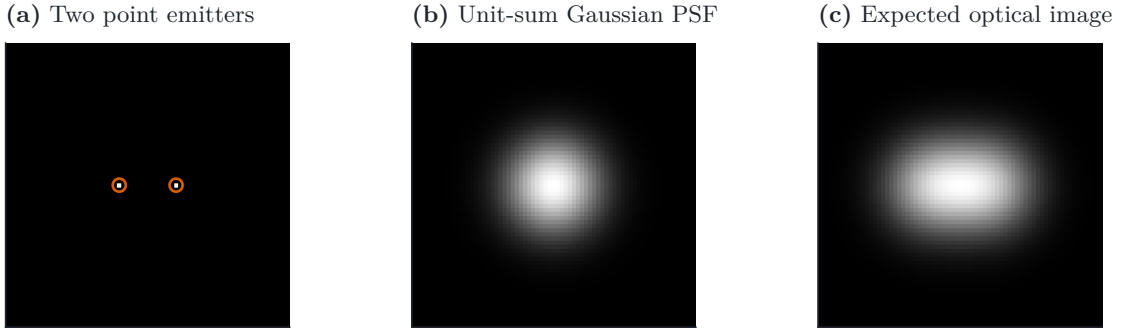
with G_y defined the same way along the rows. The factor G_x is simply the fraction of the Gaussian's light that falls within the width of pixel j ; erf is the error function. Equations (7) and (8) give the *forward model*: they say what the camera *should* record, on average, for an emitter at (x_0, y_0) . Localization, in section 7, is the inverse problem of estimating the emitter position from the recorded image.

5 Photon Noise

Light arrives as discrete photons at random times. If the expected count in a pixel is M_{ij} , the number it *actually* records, Z_{ij} , fluctuates according to Poisson statistics [8]:

$$Z_{ij} \sim \text{Poisson}(M_{ij}). \quad (9)$$

A defining property of the Poisson distribution is that its variance equals its mean. Real cameras add a second source of noise, electronic noise (*read noise*) to the photon noise, so the total



$g = f * h$; camera pixels integrate g over their area. Images scaled separately for visibility.

Figure 3: Linear shift-invariant, incoherent image formation: two equal point sources convolved with a normalized Gaussian PSF. Pixel integration is a subsequent detector operation. The brightness in each panel is scaled independently; the circles identify source positions.

per-pixel variance is approximately [14]

$$\text{Var}[Z_{ij}] \approx M_{ij} + \left(\frac{\sigma_{\text{read}}}{g} \right)^2, \quad (10)$$

where σ_{read} is the camera read noise and g its gain.

Photon noise imposes a fundamental *precision limit* (section 7.2). The more photons an emitter delivers, the smaller the relative fluctuations, and the more precisely its position can be estimated. Figure 4 shows how repeated measurements of the same emitter scatter because of this noise. In ClassroomSTORM, the **Brightness** control sets the photon budget N and **Read noise** sets σ_{read} , so students can observe how precision improves as they increase brightness.

6 Blinking and Single-Molecule Localization Logic

The central idea is to separate nearby emitters in time. Suppose a sample contains many emitters packed more closely than the diffraction limit. If they all glow at once, their PSFs overlap and eq. (6) produces an unresolvable blur because of the diffraction limit (fig. 5a).

But suppose instead that the emitters *blink*: at any instant only a small, random subset is switched on, and those that are on happen to be far apart. Then each frame contains only a few *isolated* spots (fig. 5b). Each isolated spot can be located precisely (section 7). Over thousands of frames, different emitters become active, and the accumulated set of locations traces out the whole structure.

This is single-molecule localization microscopy. STORM [2] and PALM [3], both introduced in 2006, use photoswitchable dyes and fluorescent proteins to make molecules blink; the same logic underlies the broader field of far-field optical nanoscopy [5, 14]. The diffraction limit is never actually *broken*: it is *sidestepped*, by arranging that two unresolved emitters are never on at the same time. (Other super-resolution methods, such as STED [4], take a different route by physically sharpening the PSF.)

ClassroomSTORM reproduces this logic. The *Virtual Experiment* builds exactly such blinking videos: you choose how many emitters may be on per frame and how far apart they must be. The companion macroscopic experiment uses blinking LEDs in place of fluorophores: emitters that are large in physical size but produce data with the *same statistical structure* as SMLM. Low-cost real SMLM instruments built in the same spirit have been described in the literature [19, 20].

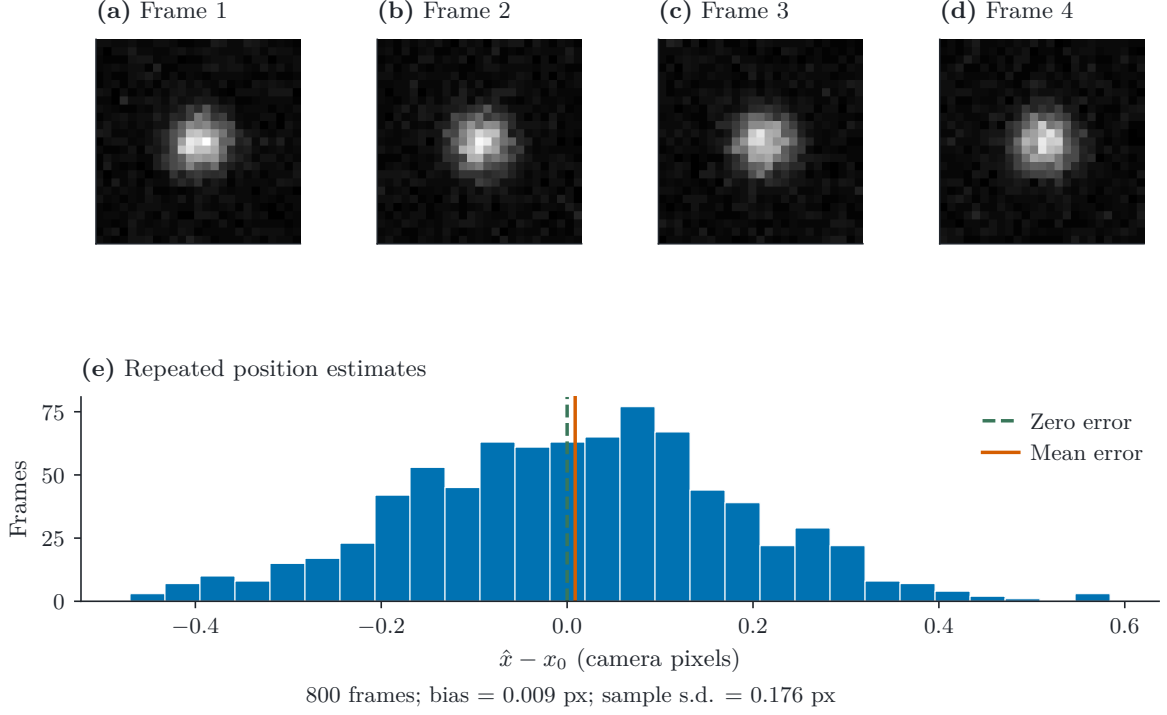
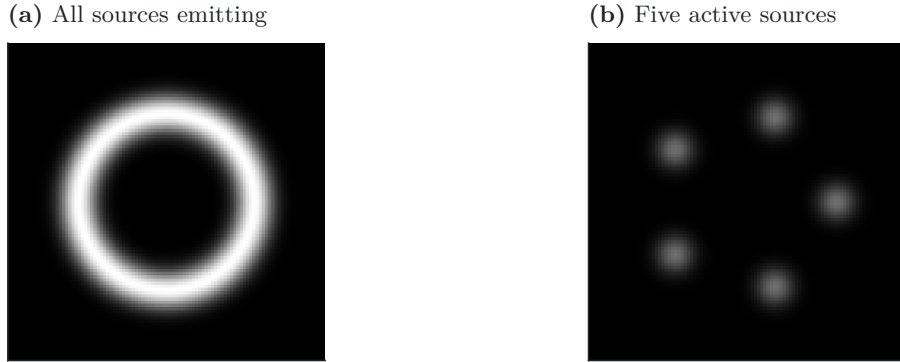


Figure 4: Pixel-integrated Gaussian signal with 2000 expected photons, $\sigma = 2.3$ pixels, and a Poisson background with a mean of 3 counts/pixel. The V1.2 single-emitter centroid uses $q = 0.98$ and frame-median subtraction. The histogram shows the spread and bias of repeated estimates, not per-event uncertainty.



Same source positions, PSF and intensity scale; only the active subset changes.

Figure 5: Temporal sparsity: the same emitter arrangement with all 40 sources active or five well-separated sources active. These are noise-free expectations with one shared intensity scale; sparse activation enables individual localization.

Why separation matters

Localization works *only* when spots are isolated. If two emitters are on together and their PSFs overlap, the methods in section 7 return one position somewhere between them, giving one incorrect position rather than two correct positions. This is why the *Virtual Experiment* lets you control `Max blinkers/frame` and `Blinker spacing`, and why crowded data reconstruct poorly.

7 Localizing an Emitter

Given one isolated, blurred spot, where is its centre? This section gives the simple method ClassroomSTORM V1.2 uses, the precision that method can achieve, and, as clearly marked background, the statistically optimal approach.

7.1 The centroid estimator used in V1.2

ClassroomSTORM V1.2 localizes with the *intensity-weighted centroid*, the “balance point” of the bright pixels [10]. First, the frame is *thresholded*: only pixels brighter than a high percentile of the frame (the top 0.5% by default) are kept, forming a mask $\mathcal{M}$ of candidate spot pixels. The centre estimate is then the brightness-weighted average of the masked pixel coordinates:

$$\hat{x}_0 = \frac{\sum_{(i,j) \in \mathcal{M}} x_j Z_{ij}}{\sum_{(i,j) \in \mathcal{M}} Z_{ij}}, \quad \hat{y}_0 = \frac{\sum_{(i,j) \in \mathcal{M}} y_i Z_{ij}}{\sum_{(i,j) \in \mathcal{M}} Z_{ij}}. \quad (11)$$

The centroid has two advantages for teaching: it needs no iterative fitting, and students can inspect every step: thresholding, masking, and weighted averaging (fig. 6). The estimate $(\hat{x}_0, \hat{y}_0)$ is a *continuous* number: it is not restricted to whole-pixel positions, so it can locate a spot to a fraction of a pixel even though the spot itself is several pixels wide.

A note on the V1.2 centroid

By default, eq. (11) uses the raw nonnegative pixel values Z_{ij} . An optional frame-median correction replaces these weights by $\max(Z_{ij} - \text{median}(Z), 0)$ before thresholding and centroiding. The median is computed over the cropped frame and assumes sparse bright emitters on a roughly uniform background. It is not a local annular estimate or a camera calibration. Background gradients and dense emitters can still bias positions; model-based alternatives are discussed in section 12.

7.2 Localization precision (background)

Because of photon noise (section 5), repeating the measurement gives a slightly different $\hat{x}_0$ each time. The spread of those estimates is the *localization precision*, written here as the standard deviation of a single position estimate, SE (not the standard error of the mean of repeated estimates). A widely used closed-form estimate is the Thompson-Larson-Webb formula [6]:

$$\text{SE} \approx \sqrt{\frac{\sigma^2 + a^2/12}{N} + \frac{8\pi \sigma^4 b_{\text{rms}}^2}{a^2 N^2}}, \quad (12)$$

where σ is the PSF width, a the pixel size, N the number of signal photons, and b_{rms} the RMS background noise per pixel in photon-equivalent units, distinct from the mean background b in the forward model. Both σ and a use the same length unit. Camera gray levels are not photon

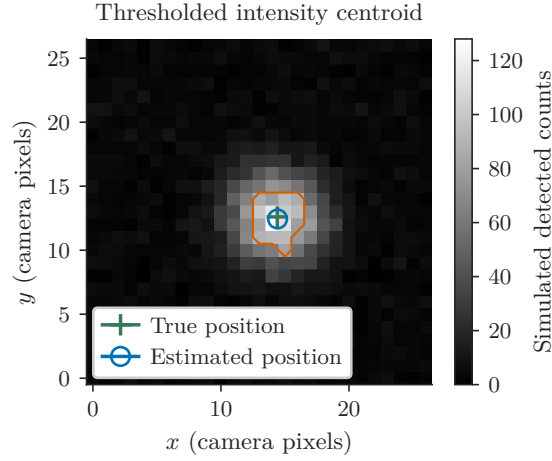


Figure 6: A synthetic pixel-integrated Gaussian spot with Poisson noise. The orange contour marks the $q = 0.98$ threshold mask; the plus and circle distinguish truth and the application centroid. Subpixel coordinates do not imply zero error. This example uses no background correction.

counts without a gain and noise calibration. This approximation is a theoretical reference, not an uncertainty estimate reported by the thresholded centroid. The formula has two terms:

- the **photon term** $(\sigma^2 + a^2/12)/N$ dominates when the signal is strong; precision improves as $1/\sqrt{N}$;
- the **background term** $8\pi\sigma^4 b_{\text{rms}}^2/(a^2 N^2)$ dominates when the signal is weak relative to the background.

Figure 7 plots this behaviour.

Worked example: resolution is not precision

Take $\sigma = 1.5$ px and a bright spot, $N = 10^4$ photons, with negligible background. The photon term gives $\text{SE} \approx \sqrt{(1.5^2)/10^4} \approx 0.015$ px, about one part in a hundred of the blur width. With only $N = 100$ photons, $\text{SE} \approx 0.15$ px. The emitter is blurred over several pixels, yet its *centre* is known to a tiny fraction of a pixel. This distinction underlies localization microscopy: *resolution* (the blur width, eq. (2)) and *precision* (how well one centre is known, eq. (12)) are different quantities, and the second can be far smaller than the first.

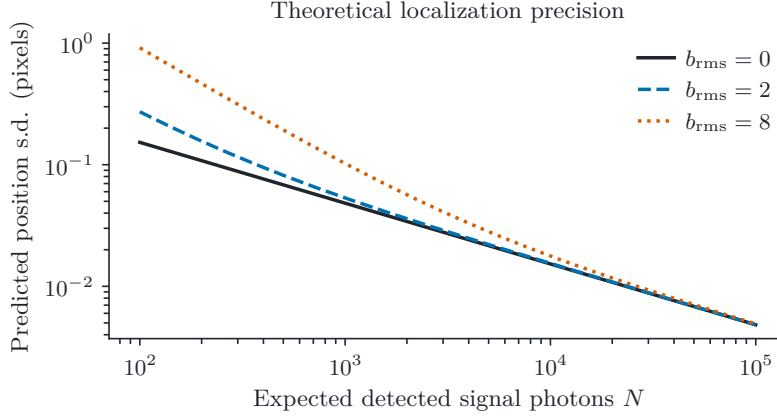
7.3 The statistical ideal (background only): MLE and the Cramér-Rao bound

The statistical ideal (background only)

The centroid of eq. (11) is transparent but can be biased by thresholding, background, or overlapping spots. A model-based alternative is maximum-likelihood estimation. For independent Poisson pixels, it minimizes

$$L(\theta) = \sum_{i,j} [M_{ij}(\theta) - Z_{ij} \ln M_{ij}(\theta)], \quad (13)$$

up to terms independent of $\theta = (x_0, y_0, N, b, \sigma)$. Under the usual regularity conditions, an



$s = 1.5$ px; pixel pitch $a = 1$ px.
 b_{rms} : RMS background noise (photon-equivalent).

Figure 7: The Thompson-Larson-Webb approximation is a theoretical reference, not an uncertainty estimate reported by the centroid software. The PSF width is $\sigma = 1.5$ camera pixels, and the pixel pitch is $a = 1$ in the same length units. b_{rms} is the RMS background noise, not the mean background. The standard deviation of position estimates scales as $N^{-1/2}$ in the signal-limited regime and N^{-1} in the background-limited regime.

unbiased estimator obeys the Cramér-Rao covariance bound [8, 7]:

$$\text{Cov}(\hat{\theta}) \succeq F^{-1}, \quad F_{mn} = \sum_{i,j} \frac{1}{M_{ij}} \frac{\partial M_{ij}}{\partial \theta_m} \frac{\partial M_{ij}}{\partial \theta_n}. \quad (14)$$

The matrix inequality means that the covariance difference is positive semidefinite. Maximum-likelihood estimators can approach the bound in an appropriate large-sample regime with the correct model; finite-sample attainment is not guaranteed. A biased thresholded centroid is not generally covered by this simple unbiased bound. V1.2 implements neither this optimizer nor per-event Fisher-information uncertainty. These equations describe a possible extension, not a fitted stage in the present pipeline.

8 Precision versus Accuracy

Two estimates can be tightly clustered yet systematically wrong. It is essential to keep two ideas separate [9]:

Precision is *repeatability*, or how closely repeated measurements of the same emitter agree with one another. It is the random scatter, and it is what eq. (12) estimates.

Accuracy is *correctness*, or how close the estimates are to the *true* position. A constant offset (a bias) reduces accuracy without reducing precision.

Figure 8 contrasts the two. The background bias of the V1.2 centroid (section 7.1) is an accuracy problem: it can shift estimates without widening their scatter. This is why ClassroomSTORM’s *Virtual Experiment* saves the true positions in `truth.csv`: comparing reconstructed positions against the truth measures *accuracy*, which precision alone cannot reveal.

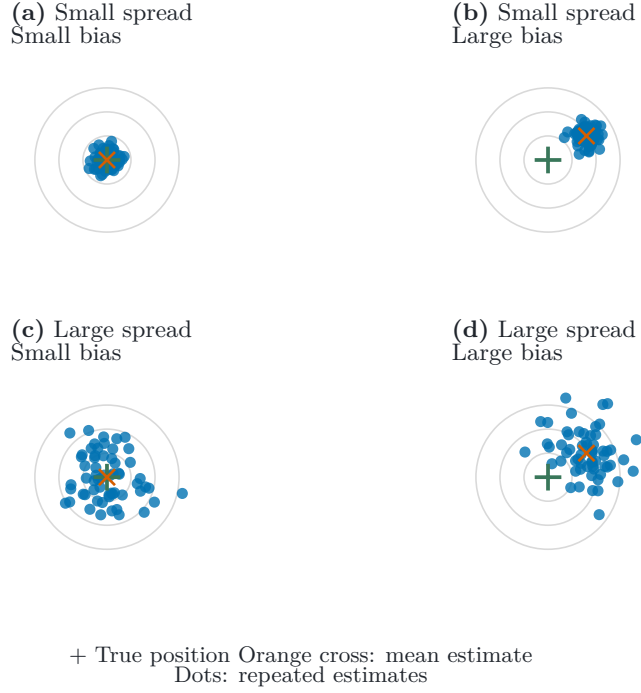


Figure 8: Illustrative samples separate repeatability (spread) from systematic offset (bias). The mean is explicitly controlled. Accuracy depends on both bias and random error; small spread alone is not accuracy.

9 From Localizations to a Reconstructed Image

A list of localized positions is not yet a picture. ClassroomSTORM *renders* it: each accepted localization is drawn as a small Gaussian “splat,” and all the splats are summed into the super-resolution image. In V1.2 every splat is drawn with the same fixed width (about 1.5 px); the splat width is a display choice that makes the sparse points visible, not a measurement. The Gaussian is evaluated at the fractional localization coordinate without rounding its centre to an integer pixel. Each displayed kernel is normalized to unit sum, including at a boundary. The output grid retains the camera pixel spacing; neither the display width nor its sampling establishes resolution.

Spatial coverage is independent of position precision. For an ideal, regularly sampled, band-limited signal with smallest spatial period d , the Nyquist condition is

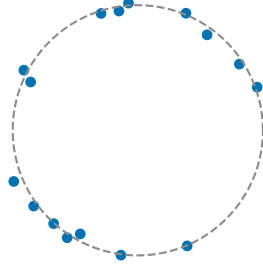
$$\Delta_s \leq d/2, \quad (15)$$

where Δ_s is the sampling pitch, not the localization uncertainty. Irregular SMLM labeling requires additional care: repeated blinking can revisit the same site without filling gaps, and an average nearest-neighbor distance alone does not guarantee adequate coverage [18, 14]. Figure 9 illustrates coverage at fixed positional noise; it does not establish a universal resolution criterion.

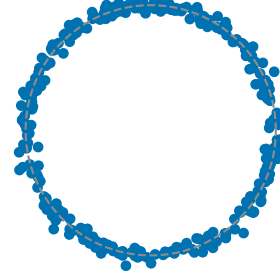
Reporting resolution

The width of a rendered splat is *not* the resolution of the result. Good reporting practice [18] is to report the localization precision (eq. (12)) and the sampling density (eq. (15)) separately, rather than quoting a single number. ClassroomSTORM is designed to make both visible rather than to report a single summary value.

(a) Sparse spatial sampling
16 positions



(b) Denser spatial sampling
320 positions



Same positional noise; different spatial coverage.
Repeated detections can revisit the same site.

Figure 9: Synthetic ring sampled at 16 or 320 independently drawn positions with the same radial noise. Sampling spacing and localization precision are different quantities. No universal resolution threshold or adequate sample count is inferred from this illustration.

10 The ClassroomSTORM V1.2 Pipeline

The list below describes what ClassroomSTORM V1.2 does when you click *Run Reconstruction* (fig. 10). The steps connect the procedure to the equations above.

1. **Load** the blinking video frame by frame (optionally cropped).
2. **Preprocess** with optional frame-median background subtraction; keep the uncorrected frames for the raw mean. Constant frames yield no localizations.
3. **Threshold** each frame: pixels brighter than a high intensity percentile (the top 0.5% by default) form the candidate mask $\mathcal{M}$.
4. **Localize** with the intensity-weighted centroid of eq. (11), once per frame in **single** mode, or once per separated bright region in **multiple** mode.
5. **Accumulate** the localizations from all frames. Retain crop-local and original-frame coordinates. If calibration is supplied, multiply original-frame coordinates by the physical scale and export the calibrated columns and unit label.
6. **Render** each localization as a Gaussian splat ($\sigma \approx 1.5$ px) and sum them into the super-resolution image.
7. **Average** the frames separately to form the diffraction-limited *raw mean* image.
8. **Compare** the raw mean and the reconstruction, using, for example, the linked profile viewer.

The *linked profile viewer* lets you compare the two images using profiles through a chosen point. Writing $I(x, y)$ for an image and (x_0, y_0) for the chosen point, the viewer plots the row and column profiles

$$I_x(x) = I(x, y_0), \quad I_y(y) = I(x_0, y), \quad (16)$$

for both the raw mean and the reconstruction. These profiles compare the displayed images. Their widths and valley depths depend on the rendering kernel and normalization, so narrow reconstructed peaks alone do not establish resolution. Figure 11 instead uses known source positions and a reproducible synthetic run; position errors are assessed against that truth.

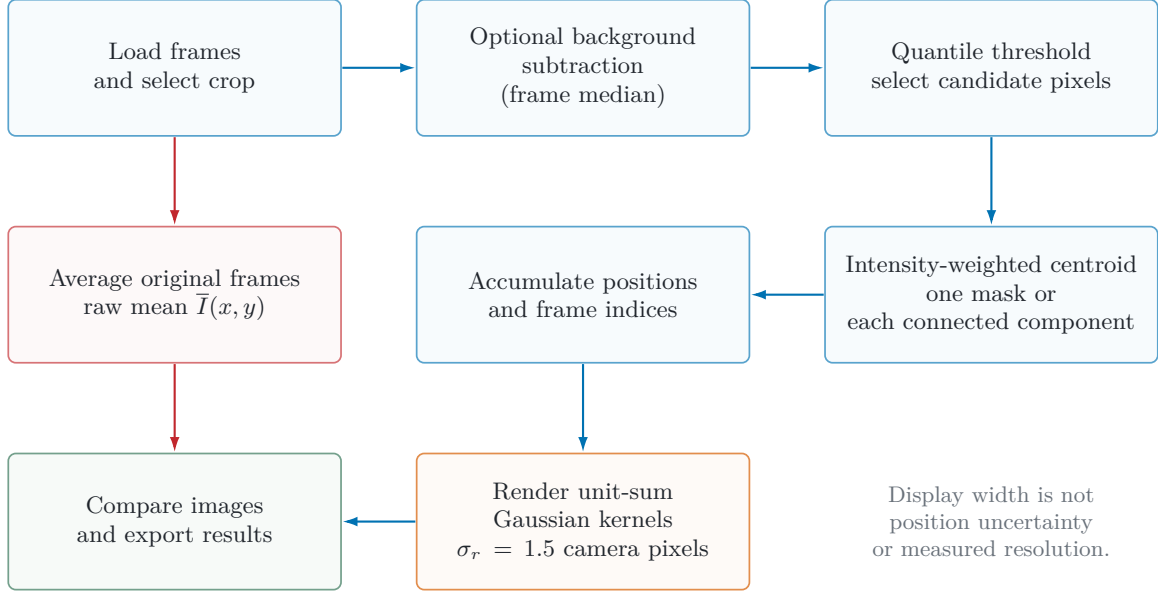
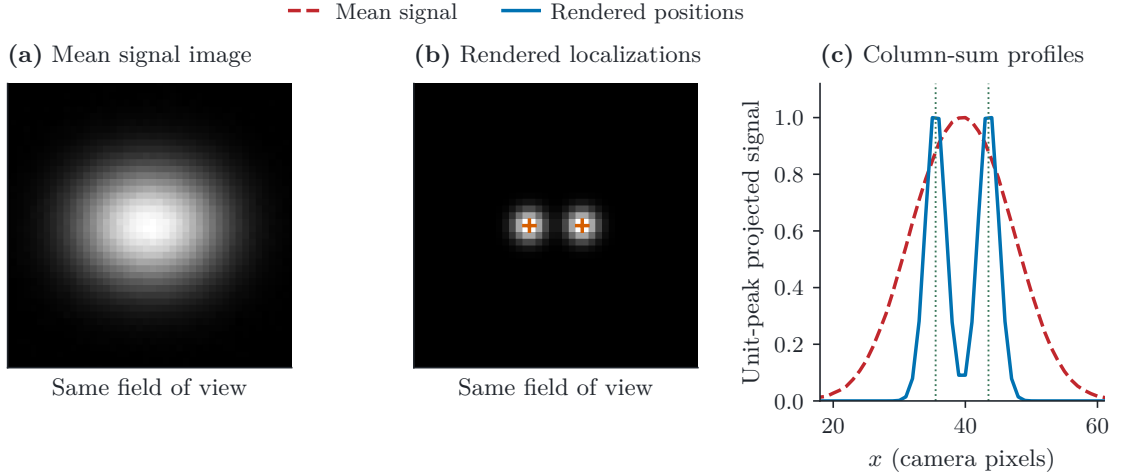


Figure 10: The current centroid pipeline, with a separate raw-mean branch. The rendering width is fixed and is not an uncertainty estimate.



Synthetic sparse blinking; same camera grid. Render width is a display parameter.

Figure 11: Actual V1.2 centroid reconstruction of 600 synthetic pixel-integrated Poisson frames. Exactly one of two sources is active per frame. The separation is 8 pixels, and the optical Gaussian width is $\sigma = 6$ pixels. The threshold is $q = 0.995$, with frame-median subtraction. The known simulated background is subtracted from the mean image for display. Both images and column-sum profiles are independently peak-normalized. Green guides and orange crosses mark the true positions. The Gaussian rendering width of $\sigma_r = 1.5$ pixels does not measure resolution. Source data and run parameters accompany the figure.

11 Scope, Assumptions, and Limitations

ClassRoomSTORM V1.2 is a teaching tool. Each limitation below provides a topic for student discussion.

What ClassRoomSTORM V1.2 does and does not do

- **Overlapping emitters cannot be separated.** If two spots merge, the centroid of the merged region returns a single incorrect position (section 6). Reliable reconstruction needs sparse, separated blinking.
- **The centroid gives a position but no uncertainty.** V1.2 does not compute a per-localization standard error. Equation (12) tells you what precision to *expect*; the software does not report it spot by spot.
- **Limited background correction.** Optional frame-median subtraction assumes sparse emitters and does not model gradients or camera-specific noise.
- **No model fitting.** V1.2 does not perform weighted-least-squares or maximum-likelihood Gaussian fitting; it does not evaluate the Cramér-Rao bound.
- **The render width is fixed.** Every splat is drawn at the same width; it shows where localizations are, not the precision of each localization.
- **No drift correction.** Optional physical calibration rescales exported coordinates but does not correct motion or estimate precision.
- **Noisy, saturated, or underexposed video reconstructs poorly,** because thresholding (section 5) can no longer separate signal from noise.

These are deliberate simplifications that make every step easy to follow. They are the starting points for the extensions below.

12 Extensions and Further Reading

The V1.2 method can be improved in well-understood ways; these are standard in research SMLM and can serve as project extensions. They are *not* part of ClassRoomSTORM Studio V1.2.

- **Local background estimation,** for example an annular median, can extend the current frame-median option when its assumptions fail.
- **Gaussian fitting.** Replacing the centroid with a weighted least-squares or maximum-likelihood fit of the model in eq. (7) approaches the Cramér-Rao bound (eq. (14)) and yields a per-localization precision [7, 10].
- **Drift correction.** Slow stage drift can be measured by cross-correlating frames and subtracted before rendering [12].
- **Resolution metrics.** Fourier ring correlation gives an image-based resolution estimate from two independent halves of the data [18].
- **Reference software.** Research packages such as ThunderSTORM [11] implement the full pipeline and are useful for comparison. An accessible introduction to SMLM data analysis is given by Lee *et al.* [17], and the mathematical theory of recovering point sources is treated by Candès and Fernández-Granda [22].

A A Short Derivation: Why Precision Scales as $1/\sqrt{N}$

Derivation (optional)

A localization is, in essence, an average of the positions where the N signal photons arrived. Each photon lands at a random position drawn from the PSF, which has a spread of order σ . The standard error of the mean of N independent samples, each with spread σ , is

$$\text{SE} \approx \frac{\sigma}{\sqrt{N}}.$$

This is the leading (photon-limited) term of the Thompson formula, eq. (12); the $a^2/12$ correction accounts for the finite pixel size, and the second term accounts for background photons that carry no position information. The lesson is simple and general: *collect more photons and the position estimate becomes more precise, in proportion to the square root of the count.*

B Symbol Table

Symbol	Meaning
λ	Wavelength of light
NA	Numerical aperture of the imaging lens
$d_{\text{Abbe}}, d_{\text{Rayleigh}}$	Diffraction-limited resolution (eqs. (2) and (3))
h, f, g	Point spread function, true scene, recorded image (eq. (6))
σ	Width of the (Gaussian) point spread function
(x_0, y_0)	True position of an emitter; $(\hat{x}_0, \hat{y}_0)$ is its estimate
N	Number of signal photons from an emitter
b	Background level per pixel
a	Camera pixel size
M_{ij}	Expected counts in pixel (i, j) (eq. (7))
Z_{ij}	Recorded counts in pixel (i, j)
σ_{read}, g	Camera read noise and gain
$\mathcal{M}$	Set of thresholded (candidate spot) pixels
SE	Single-position standard deviation
$L(\theta), F$	Negative log-likelihood and Fisher information (eqs. (13) and (14))

Table 1: Symbols used in these notes.

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